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Giuseppe Orlando

Curriculum Vitae

Personal information

Born in GAGLIANO DEL CAPO (LE), ITALY, 18/10/1994
Citizenship: ITALIAN

Education - Work experience

- 01/10/2023 - Present: **Postdoctoral researcher**
  Institution: École polytechnique (Palaiseau, France)
  Laboratory and group: CMAP (Centre de Mathématiques Appliquées), HPC@Maths group
  Funding: OPEN NUM DEF project (PI Prof. Marc Massot, Dr. Loïc Gouarin, Dr. Teddy Pichard)
  Supervisor: Prof. Marc Massot
  Research topic: Development of numerical methods for liquid-gas two-phase flows with multi-scale modelling.
- 01/02/2023 - 30/09/2023: **Research fellowship (post-doc)**
  Institution: Politecnico di Milano (Milan, Italy)
  Laboratory: MOX - Department of Mathematics
  Funding: ESCAPE-2 H2020, projet Grant Agreement No. 800897
  Supervisor: Prof. Luca Bonaventura
  Research topic: Development of discontinuous finite elements codes for atmosphere dynamics simulations in the framework of `deal.II` library.
- 01/11/2019 - 31/01/2023: **Phd candidate** for the programme MATHEMATICAL MODELS AND METHODS IN ENGINEERING
  Institution: Politecnico di Milano (Milan, Italy)
  Laboratory: MOX - Department of Mathematics
  Supervisor: Prof. Luca Bonaventura
  Co-supervisor: Prof. Paolo Francesco Barbante
  Research project: Modelling and simulations of two-phase flows including geometric variables
  Defence: 18/04/2023, Milan, Italy
  Commission:

Prof. Davide Carlo Ambrosi	Politecnico di Torino	President
Prof. Simone Scacchi	Università degli Studi di Milano	Member
Prof. Luca Formaggia	Politecnico di Milano	Member

- 29/11/2021 - 15/07/2022: **Visiting student at CMAP** (Centre de Mathématiques Appliquées), École polytechnique (Palaiseau, France)
- 2016 - 2019: **Master of Science Degree** in MATHEMATICAL ENGINEERING
  Track: Computational Science and Engineering
  Institution: Politecnico di Milano (Milan, Italy)
  Thesis: Development of a reactive numerical solver based on the SU2 CFD code (discussion 03/10/2019)
  Supervisor: Prof. Paolo Francesco Barbante
  Mark: 110/110
- 2013 - 2016: **Bachelor of Science Degree** in MATHEMATICAL ENGINEERING
  Institution: Politecnico di Milano (Milan, Italy)
  Thesis: Equazioni ellittiche degeneri strutturate su campi di Hörmander (discussion 22/09/2016)
  Supervisor: Prof. Marco Bramanti
  Mark: 107/110

- 2013: **Diploma**: General High School
  Institution: Istruzione di Istituto Superiore G. Stampacchia (Tricase (LE), Italy)
  Mark: 100L/100

Training Programmes

- 2018: Introductory course on **GPU** computing for Fluid Dynamics applications
  `https://github.com/GiuseppeOrlando878776/MilanoGPU2018`
- 2019: Crash course on Vehicle Aerodynamics

Technical skills

- Programming languages
  - **C (MPI), C++, MATLAB, FreeFem++, Python**: Academic level with the development of small projects
    `https://github.com/GiuseppeOrlando878776/Progetto_ANEDP`
    `https://github.com/GiuseppeOrlando878776/TwoPhase_Flow`
    C++ code for incompressible Navier-Stokes equations included in the official `deal.II` code-gallery
    (`https://dealii.org/developer/doxygen/deal.II/code_gallery_NavierStokes_TRBDF2_DG.html`)
    Development of codes in the open-source framework `samurai`
    (`https://github.com/hpc-maths/2025_09_two_fluid_all_topology`)
  - R, Mathematica: Basic level
- Other
  Jupyter, **Paraview**: Academic level

Language skills

- Italian: Mother language
- British English: B2 Level, TOEFL iBT Certificate, Mark 86/120
- French: Written: fair, Spoken: fair

Computer skills

- Operating systems
  Windows, **Linux**
- Text and spreadsheets
  Microsoft Office, **LaTeX**

Research interests

- Main research interests: Computational Fluid Dynamics (**CFD**), Discontinuous Galerkin (**DG**) methods, Two-phase flows, low Mach flows, **IMEX** time discretization methods, **variational** modelling
- Other research interests: Monotonization techniques, Numerical Weather Predictions (**NWP**), shallow water (**SW**) equations, lava flow models, Scientific Machine Learning (**SciML**)

Outline of research activity

My research activity has been mainly focused on the field of Computational Fluid Dynamics (**CFD**), spanning a wide range of applications, starting from my Master thesis on the development of a numerical solver for reactive flows, with a particular emphasis on approaches which are robust in the case of vanishing species.
During my PhD studies [21], I started to focus on the modelling and simulations of **two-phase flows**, which represents my actual main research topic. In particular, I took into account geometric variables, such as the interfacial area density, whose accurate computation is a prerequisite in order to obtain reliable values of phase exchange terms [7], and the interface mean curvature [16]. This allowed me to spend some months abroad and to start a collaboration with Centre de Mathématiques Appliquées (CMAP) at École polytechnique, where I am actually a postdoctoral researcher.
My first publication [12] deals with the development of an efficient numerical scheme for the incompressible Navier-Stokes equations. The proposed approach combines flexible discontinuous finite element space discretizations with efficient semi-implicit time discretization methods and employs an artificial compressibility formulation in order to overcome some well known issues of projection methods. More recently, an extension of this approach to two-phase flows has been proposed [6].
The first significant contribution to the literature is presented in [11] and represents a non straightforward extension to the compressible Navier-Stokes equations of real gases of **IMEX** approaches originally employed for ideal gases. More in detail, a suitable fixed point loop has been introduced for the general cubic equation of state, which can effectively handle experimental data. A discontinuous Galerkin (**DG**) spatial discretization has been employed, so as to easily achieve a full second-order method. Moreover, suitable refinement indicators have been introduced for non-ideal gases, so as to effectively use Adaptive Mesh Refinement (**AMR**). An analysis of the employed second order IMEX-RK method has been carried out, providing an improvement in terms of absolute monotonicity.
I also focused on the application of the IMEX scheme proposed in [11] to atmosphere dynamics and Numerical Weather Prediction (**NWP**) [9], showing the effectiveness of the method in a number of test cases even with non-trivial and real orography profiles. The efficiency of the proposed method and solver have been demonstrated in [14, 15]. Moreover, for the first time in the literature, the robustness of the method using non-conforming meshes [5] and curved elements [4] for flows over orography has been proven. Finally, an analysis of a general class of IMEX-RK methods in terms of asymptotic-preserving (**AP**) property has been recently performed [3] and higher order methods with respect to that employed in [11] have been validated [2, 3].

In addition, I developed a filtering monotonization technique [10] in order to avoid (or at least reduce) the oscillations that arise when high order discretization methods are used. Its potentialities have been shown in a number of benchmarks, including test cases for the Euler equations of gas dynamics.

Furthermore, I also tested for the first time in the literature to use **language models** for the development of solvers for PDEs based on the finite element method [8].

Nowadays, I am mainly working on the modelling and simulations of **two-phase flows**, with particular emphasis on the derivation of models through **variational** principles [1, 13, 19] and on the analysis of a hierarchy of Baer-Nunziato type models for compressible two-phase flows [17]. Furthermore, I am working on the development of robust and accurate models for the shallow water (de Saint-Venant) equations [20], including lava flow models [18].

Scientific publications

List of publications

[1] W. Haegeman et al. "An all-topology two-fluid model for two-phase flows derived through Hamilton's Stationary Action Principle". *Proceedings of the Royal Society A* (2025). (in press. Preprint: `https://hal.science/hal-05249139`).

[2] G. Orlando, S. Boscarino, and G. Russo. "A quantitative comparison of high-order asymptotic-preserving

and asymptotically-accurate IMEX methods for the Euler equations with non-ideal gases". *Computer Methods in Applied Mechanics and Engineering* 442 (2025), p. 118037. DOI: 10.1016/j.cma.2025.118037.

[3] G. Orlando and L. Bonaventura. "Asymptotic-preserving IMEX schemes for the Euler equations of non-ideal gases". *Journal of Computational Physics* 529 (2025), p. 113889. DOI: 10.1016/j.jcp.2025.113889.

[4] G. Orlando, T. Benacchio, and L. Bonaventura. "Impact of curved elements for flows over orography with a Discontinuous Galerkin scheme". *Journal of Computational Physics* 519 (2024), p. 113445. DOI: 10.1016/j.jcp.2024.113445.

[5] G. Orlando, T. Benacchio, and L. Bonaventura. "Robust and accurate simulations of flows over orography using non-conforming meshes". *Quarterly Journal of the Royal Meteorological Society* 150 (2024), pp. 4750–4770. DOI: 10.1002/qj.4839.

[6] G. Orlando. "An implicit DG solver for incompressible two-phase flows with an artificial compressibility formulation". *International Journal for Numerical Methods in Fluids* 96 (2024). Preprint: https://arxiv.org/abs/2307.04580, pp. 1932–1959. DOI: 10.1002/fld.5328.

[7] G. Orlando, P.F. Barbante, and L. Bonaventura. "On the evolution equations of interfacial variables in two-phase flows". *International Journal of Multiphase Flow* 169 (2023), p. 104617. DOI: 10.1016/j.ijmultiphaseflow.2023.104617.

[8] G. Orlando. "Assessing ChatGPT for coding finite element methods". *Journal of Machine Learning for Modeling and Computing* (2023). DOI: 10.1615/JMachLearnModelComput.2023049326.

[9] G. Orlando, T. Benacchio, and L. Bonaventura. "An IMEX-DG solver for atmospheric dynamics simulations with adaptive mesh refinement". *Journal of Computational and Applied Mathematics* (2023), p. 115124. DOI: 10.1016/j.cam.2023.115124.

[10] G. Orlando. "A filtering monotonization approach for DG discretizations of hyperbolic problems". *Computers & Mathematics with Applications* 129 (2023). Preprint: https://arxiv.org/abs/2204.08693, pp. 113–125. DOI: 10.1016/j.camwa.2022.11.017.

[11] G. Orlando, P.F. Barbante, and L. Bonaventura. "An efficient IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases". *Journal of Computational Physics* 471 (2022), p. 111653. DOI: 10.1016/j.jcp.2022.111653.

[12] G. Orlando et al. "An efficient and accurate implicit DG solver for the incompressible Navier-Stokes equations". *International Journal for Numerical Methods in Fluids* 94 (2022), pp. 1484–1516. DOI: 10.1002/fld.5098.

List of proceedings

[13] W. Haegeman et al. "A generic framework to derive systems of conservation laws with source terms and its application to heat conduction in fluid flows: An alternative to the method of moments in kinetic theory of gases?" *ESAIM: Proceedings and Surveys* 78 (2025), pp. 80–97. DOI: 10.1051/proc/202578080.

[14] G. Orlando, T. Benacchio, and L. Bonaventura. "Improving the scalability of a high-order atmospheric dynamics solver based on the deal.II library". *Procedia Computer Science* 267 (2025). Proceedings of the Third EuroHPC Day, pp. 227–236. DOI: 10.1016/j.procs.2025.08.249.

[15] G. Orlando, T. Benacchio, and L. Bonaventura. "Efficient and scalable atmospheric dynamics simulations using non-conforming meshes". *Procedia Computer Science* 255 (2025). Proceedings of the Second EuroHPC Day, pp. 33–42. DOI: 10.1016/j.procs.2025.02.258.

[16] G. Orlando, P. F. Barbante, and L. Bonaventura. "Assessment of Different Strategies to Compute Surface Tension Contribution". *Numerical Mathematics and Advanced Applications ENUMATH 2023, Volume 2: European Conference, September 4-8, Lisbon, Portugal*. Vol. 154. Preprint: https://arxiv.org/abs/2402.04670. Springer Nature. 2025, pp. 213–222. DOI: 10.1007/978-3-031-86169-7_22.

List of preprints

[17] G. Orlando et al. *A robust computational framework for the mixture-energy-consistent six-equation two-phase model with instantaneous mechanical relaxation terms*. 2025. arXiv: 2509.26284 [math.NA]. URL: https://arxiv.org/abs/2509.26284.

[18] F. Gatti and G. Orlando. *Second-order Optimally Stable IMEX (pseudo-)staggered Galerkin discretization: application to lava flow modeling*. 2025. arXiv: 2509.09460 [math.NA]. URL: https://arxiv.org/abs/2509.09460.

[19] W. Haegeman et al. *Modelling of relative velocity, velocity fluctuations and their interactions for two-fluid models by Stationary Action Principle*. 2025. URL: https://hal.science/hal-05224712.

[20] L. Arpaia et al. *A high-order matrix-free adaptive solver for the shallow water equations with irregular bathymetry*. 2025. arXiv: 2505.18743 [cs.CE]. URL: https://arxiv.org/abs/2505.18743.

PhD thesis

[21] G. Orlando. "Modelling and simulations of two-phase flows including geometric variables". http://hdl.handle.net/10589/198599. PhD thesis. Politecnico di Milano, 2023.

Master thesis

[22] G. Orlando. *Development of a reactive numerical solver based on the SU2 CFD code*. http://hdl.handle.net/10589/150021. Master thesis, Politecnico di Milano. 2019.

Editorial activity

- 01/01/2026 - :Associate editor for *Monthly Weather Review* (*MWR*)
- I have acted as a referee for international journals in Mathematical Modelling and Scientific Computing: *Computers and Mathematics with Applications* (*CAMWA*), *Journal of Computational Physics* (*JCP*), *Computer & Fluids* (*CAF*), *Monthly Weather Review* (*MWR*), *International Journal on Geomathematics* (*GEM*).

Main conferences partecipation

- **Invited seminar** for the series "Séminaire Analyse Appliquée (AA)" at I2M (Institut de Mathématiques de Marseille), 31/03/2026, 45 minutes talk "An all-topology two-fluid model for two-phase flows derived through Hamilton's Stationary Action Principle"
- Speaker Flash workshop, 02/02/2026, 50 minutes talk "A unified compressible two-phase, two-scale model with capillarity for separated-to-disperse transitions in atomizing flows"
- **Invited speaker** at 38ème séminaire de mécanique des fluides numérique, 26/01/2026, 40 minutes talk "A unified compressible two-phase, two-scale model with capillarity for separated-to-disperse transitions in atomizing flows"
- **Invited seminar** for the series "Séminaires de Mathématiques et de leurs Applications" at LMAP (Laboratoire de Mathématiques et de leurs Applications, Pau), 22/01/2026, 45 minutes talk "An adaptive high-order DG solver for the shallow water equations with irregular bathymetry"
- **Invited speaker** at ARIA (Accurate ROMs for Industrial Applications) 2025 workshop, 23/09/2025, 30 minutes talk "A high-order matrix-free adaptive DG solver for the shallow water equations with irregular bathymetry"
- 11th International Conference on Coupled Problems in Science and Engineering (COUPLED 2025), 27/05/2025, 15 minutes talk "A high-order solver for NWP using non-conforming adaptive andcurved meshes: applications to flows over orography"
- 12th International Conference on Multiphase Flow (ICMF 2025), 14/05/2025, 15 minutes talk "A unified two-scale gas-liquid two-fluid model with capillarity and interface regularization through a mass transfer between scales"
- **Invited seminar** for the series "Séminaire de Calcul Scientifique et Modélisation" at IMB (Institut de Mathématiques de Bordeaux), 10/04/2025, 45 minutes talk "Asymptotic-preserving IMEX schemes for the Euler equations of non-ideal gases"

- **Invited speaker** at Seminar Mathematics and Atmospheric Physics, 23/10/2024, 45 minutes talk "Robust and accurate simulations of flows over orography using non-conforming variable resolution meshes"
- Seminar HPC@Maths, 18/10/2024, "Numerical analysis of a two-fluid model"
- European Meteorological Society Annual Meeting, 06/09/2024, 15 minutes talk "Robust and accurate simulations of flows over orography using non-conforming variable resolution meshes"
- 2nd BICTAM-CISM Symposium on Dispersed Multiphase Flows: from Measuring to Modeling, 22/08/2024, 15 minutes talk "A unified two-scale gas-liquid two-fluid model with capillarity and interface regularization through a mass transfer between scales"
- International Conference on Numerical Methods in Multiphase Flows - 5 (ICNMMF-5), 28/06/2024, 15 minutes talk "A two-scale model for two-phase flows including mass transfer"
- 9th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS 2024), 06/06/2024, 15 minutes talk "A two-scale model for two-phase flows including geometric variables and mass transfer"
- 46ème Congrès National d'Analyse Numérique (CANUM 2024), 29/05/2024, 15 minutes talk "A two-scale model for two-phase flows including geometric variables and mass transfer"
- **Invited speaker** at the INdAM Workshp INSiDEs (Innovations in the Numerical Treatment of Stiff Differential Equations, 22/02/2024, 45 minutes talk "An asymptotic-preserving scheme for Euler equations with non-ideal gases"
- Seminar HPC@Maths, 24/11/2023 "Derivation of relations for interfacial area density in two-phase flows through the Stationary Action Principle"
- European Conference on Numerical Mathematics and Advanced Applications (ENUMATH 2023), 05/09/2023, 25 minutes talk "A filtering monotonization technique for DG discretization of hyperbolic problems"
- Bi-annual congress of SIMAI 2023, 29/08/2023, 25 minutes talk "A filtering monotonization technique for DG discretization of hyperbolic problems"
- 17th U. S. National Congress on Computational Mechanics (USNCCM17), 25/07/2023, 20 minutes talk "An IMEX-DG solver for atmospheric dynamics with Adaptive Mesh Refinement"
- PDEs on the sphere 2023 (PDEs 2023), 06/07/2023, 30 minutes talk "An IMEX-DG solver for atmospheric dynamics based on deal.II framework"
- 7th ECCOMAS Young Investigators Conference (YIC2023), 21/06/2023, 15 minutes talk "An IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases"
- **Invited seminar** for the series "Seminario di Modellistica Differenziale Numerica" at Sapienza Univeristá di Roma, 22/11/2022, 45 mniutes talk "A filtering monotonization technique for DG discretizations of hyperbolic problems"
- 14th European Fluid Mechanics Conference (EFMC14), 14/09/2022, 12 minutes talk "An IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases"
- XVIII International Conference on Hyperbolic Problems: Theory, Numerics, Applications (HYP2022), 21/06/2022, 15 minutes talk "An IMEX-DG solver for the compressible Navier-Stokes equations with a general equation of state"
- 45ème Congrès National d'Analyse Numérique (CANUM 2020), 16/06/2022, 15 minutes talk "An IMEX-DG solver for the compressible Navier-Stokes equations with a general equation of state"
- Bi-annual congress of SIMAI 2020+2021, 30/08/2021, 25 minutes talk "An accurate and efficient implicit Discontinuous Galerkin solver for the incompressible Navier-Stokes equations"
- Conference "Numerical methods for hyperbolic problems (NumHyp 2021)", 26/07/2021, 30 minutes talk "An efficient and accurate implicit DG solver for the incompressible Navier-Stokes equations"

Outline of teaching activity

Responsible teacher

- Crash course on the use of MATLAB software for students of Lecco Campus at Politecnico di Milano - Spring Semester A.Y. 2019-2020 and 2020-2021
  Teaching load: 16 hours (per course)

Exercise Session

- APM51656EP "Elements of measure theory and functional analysis" - A.Y. 2025-2026, École polytechnique (M1)
  Teaching load: 6 hours
  Responsible of the course: Prof. David Gérard-Varet

- MECCANICA RAZIONALE (10 CFU) - A.Y. 2020-2021, Politecnico di Milano
  Teaching load: 20 hours
  Responsible of the course: Prof. Maurizio Stefano Vianello

- MATHEMATICAL MODELS AND METHODS IN ENGINEERING [1] (5 CFU) - A.Y. 2019-2020 and 2020-2021, Politecnico di Milano
  Teaching load: 10 hours (A.Y. 2019-2020), 20 hours (A.Y. 2020-2021)
  Responsible of the course: Prof. Maurizio Stefano Vianello

Informatic Laboratory

- **Informatic Laboratory** for the course SHAPES AND ALGORITHMS: TOOLS FOR GENERATIVE DESIGN (6 CFU) - A.Y. 2020-2021, Politecnico di Milano
  Teaching load: 26 hours
  Responsible of the course: Prof. Anna Scotti

Tutoring

- **Tutor of the final project** for the APM51051EP course "Systémes dynamiques pour la modèlisation et la simulation des milieux rèactifs multi-èchelles" - A.Y. 2024-2025, 2025-2026, École polytechnique (M2)
  Responsible of the course: Prof. Marc Massot

- **Tutor of the final project** for the MAP551 course "Systémes dynamiques pour la modèlisation et la simulation des milieux rèactifs multi-èchelles" - A.Y. 2023-2024, École polytechnique (M2)
  Responsible of the course: Prof. Marc Massot

Supervision activity Master thesis

1. Safira Leandra Liora Piasko (ETH-Zurich, September 2022 - March 2023)
   Title: Discontinuous Finite Element Methods for Large Eddy Simulation(LES) of Incompressible Flows
   Collaborators: Prof. Luca Bonaventura, Prof. Antonella Abbà, Prof. Christoph Schär

2. Letizia Bottani (Double program Politecnico di Milano - Technical University of Denmark, September 2025 - February 2026)
   Title: Simulation of rotating flows with a high-order finite element framework for atmospheric dynamics with adaptive mesh refinement
   Collaborators: Prof. Luca Bonaventura, Dr. Tommaso Benacchio, Prof. Allan Peter Engsig-Karup

Projects, organizations and other info

- 17/11/2025: **Co-organiser** of of the practical training session "Hands-on Samurai", dedicated to introducing the `samurai` code to approximately 25 external participants from 10 different research institutions (held on site at CMAP)

- 07/07/2025 - 01/08/2025 : Participation to Modeling Summer Visit (MSV) Program as co-PI of a research project.
  Title: A two-phase flow model with scale separation for the separated-to-disperse phase transition
  Institution: NASA Ames
  Funding: Partial support by Fondation Mathématique Jacques Hadamard (2k €) through the program Junior Scientific Visibility

- **Co-organization of seminars HPC@Maths** at CMAP - École polytechnique starting from November 2023
- **Co-organisation of a minisymposium** at SIMAI 2025 Congress
  Title: Recent advancements in numerical methods and physical modelling of multi-material flows
  Co-organizers: Niccolò Tonicello (SISSA), Federico Dalla Barba (Padua University), Marica Pelanti (ENSTA)
- 28/08/2023 - 01/09/2023: **Staff member** of Bi-annual congress of SIMAI 2023
- Solver for incompressible Navier-Stokes equations included in the **official** `deal.II` **code-gallery**
  `https://dealii.org/developer/doxygen/deal.II/code_gallery_NavierStokes_TRBDF2_DG.html`
- PI of the EuroHPC Benchmark project EHPC-BEN-2024B03-045
  Title: A high-order efficient and scalable solver for atmospheric dynamics using non-conforming and curved meshes
- PI of the EuroHPC Development project EHPC-DEV-2024D10-054
  Title: A high-order efficient and scalable solver for NWP using dynamically adaptive non-conforming meshes
- PI of the ISCRA-C Project NUMNETF
  Title: NUmerical Methods for Non-Equilibrium Two-phase Flows
  Research Group: Paolo Francesco Barbante, Luca Bonaventura
- PI of the ISCRA-C Project SIDICONS
  Title: Semi-Implicit Discretizations for Incompressible and COmpressible Navier-Stokes equations
  Research group: Antonella Abbá, Paolo Francesco Barbante, Luca Bonaventura

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